A GPGPU-Based Framework for 2-body Statistics Computation

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Abstract

Various types of two-body statistics (2-BS) are regarded as essential components for data analysis in many scientific and computing domains. However, the quadratic time complexity makes existing algorithms inefficient. Therefore, use of modern parallel hardware has become an obvious direction for research and practice on this topic. The objective of this project is designing and optimizing parallel algorithms for 2-BS computation on Graphics Processing Units (GPUs). We present modifications to a naïve algorithm by various novel techniques in the computation and data output stages of such problems. We develop models to analyze and optimize such techniques for different types of 2-BS. Experiments run on modern GPU hardware show that our GPU algorithm outperforms the best known CPU program by at least an order of magnitude in various applications. Furthermore, our algorithm achieves very high level of GPU resource utilization, indicating near-optimal performance.

2-Body Statistic

- 2-Body Statistic is a group of statistical measurements that evaluates all pairs of points among an N-point data set.
- For example: all-point nearest-neighbor classification, kernel density regression, relational joins, two-point angular correlation function, 2-point correlation function, Radial distribution function (RDF), and spatial distance histogram (SDH).
- These can be computed by solving a function between all pairs of datum. A naïve GPU algorithm is as follows:

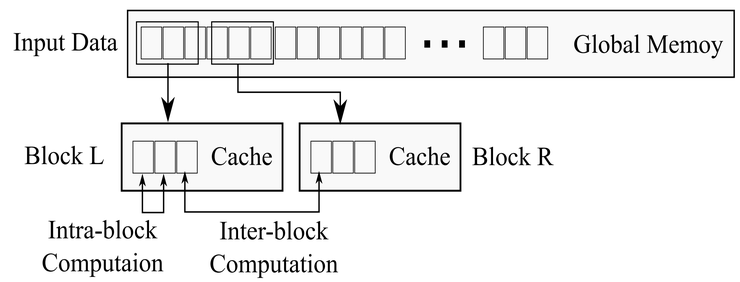
Algorithm 1: Generic GPU-based 2-BS algorithm

```
Local Var: t (Thread id)
currentPt ← input[t]
for i = t + 1 to N do
    d ← DisFunction(currentPt, input[i])
    update output with d
end for
```

- 2-Body Statistics can be classified based on the output pattern into three group:
  - **Type-I**: very small amount of output data which small enough to be placed in GPU Register.
  - **Type-II**: Medium Output which small enough to be placed in On-chip shared memory.
  - **Type-III**: Large Output that need to be placed in global memory or output size is unknown.
- 2 Stage of computation
  - Pairwise Computation Stage
  - Writing Output Stage

Computation Stage

- Divide input data into small blocks which ensures it can be put into cache.

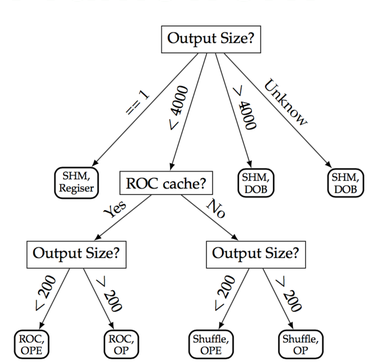


- Cache data on register, shared memory or read-only data cache depend on type of 2-Body statistic application.

Writing Output Stage

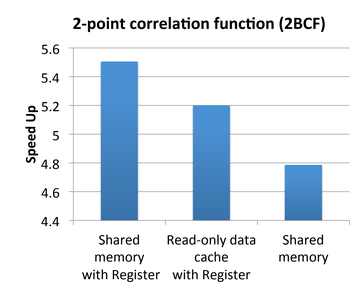
- Tiny output:
  - Using register to cache temporary output and reduction in the final phrase.
- Medium size output:
  - Shared memory is utilized with "**Advanced output privatization method**".
  - Multiple temporary output are used to maximize utilization of shared memory and minimum the update collision between threads warp level.
  - Threads on the same warp access to different copy of temporary output.
- Large size output:
  - Global memory is used with "**Direct output buffer**".
  - Handle dynamic output with only one single run of the kernel.
  - Minimum synchronization by each thread given take one page of buffer and can request new page with atomic operation.

Framework

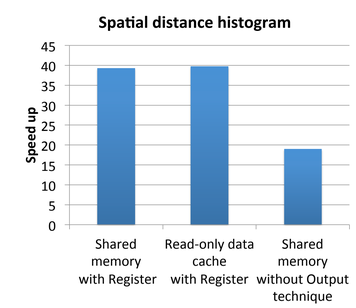


- Combine different techniques can maximize performance of GPU system.
- Framework can automatically select best technique for each application.
- It takes a small number of inputs and generates CUDA code automatically

Experimental Result



- **Type I application.**
  - 2-point correlation function.
  - Handle output by register.
  - 2 million items as input.
  - Compare speedup over generic GPU algorithm.



- **Type II application.**
  - Spatial distance histogram.
  - Handle output by Advance output privatization method.
  - 2 million items as input.
  - Compare speedup over multithread CPU algorithm.

- **Type III application.**
  - Item similarity application.
  - Handle output by Direct output buffer.
  - 2 million items as input.
  - Average 65X from multithread CPU algorithm.

Conclusion

- Existing work of 2-Body Statistic lacks a comprehensive study of the possible techniques to achieve high performance on GPUs.
- Parallel of 2-Body Statistic on GPUs is non-trivial task because the architecture of modern GPUs is much more complex than that of multi-core CPUs.
- 2-Body Statistic shared commons pattern of computation which can classify in to three type based on output pattern.
- The main computation of 2-Body Statistic contain 2-stage, which is pairwise computation stage and writing output stage.
- Each type of 2-Body Statistic require difference technique to maximize performance of GPUs.
- Our 2-Body Statistic framework can generate CUDA kernel with the best technique for that application.

References

[1] Napath Pitaksirianan, Zhila Nouri, and Yi-Cheng Tu. Efficient 2-Body Statistics Computation on GPUs: Parallelization and Beyond. In Procs. of 45th International Conference on Parallel Processing (ICPP), pp. 380-385.